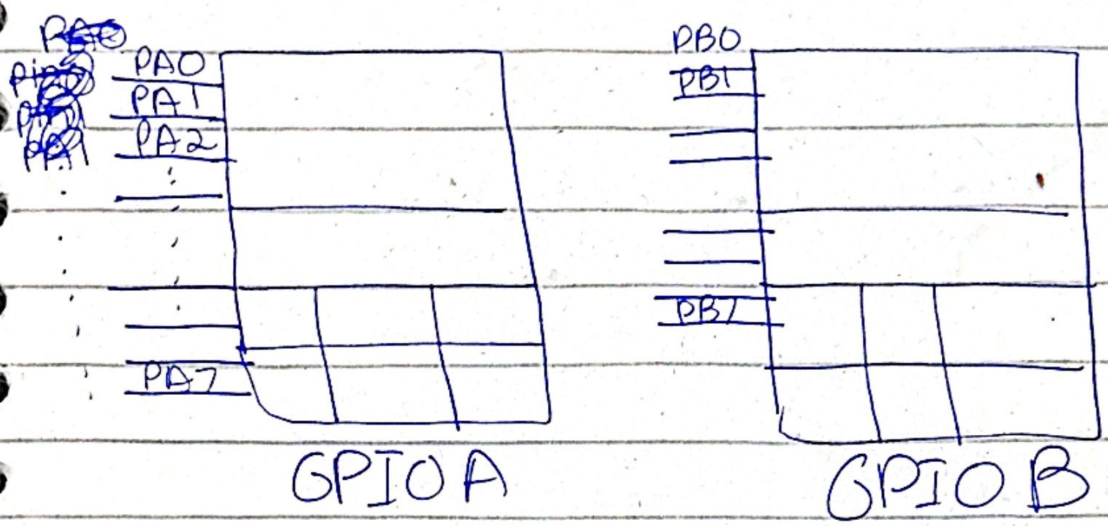
14 Oct 2025



...... GPIO F

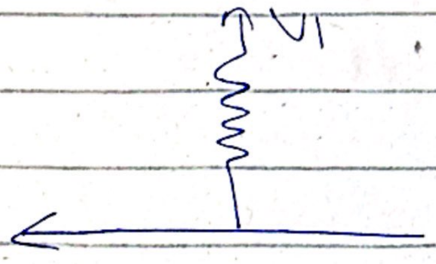
$8 \times 6 = 48$ pins

→ For higher current devices have to use isolators

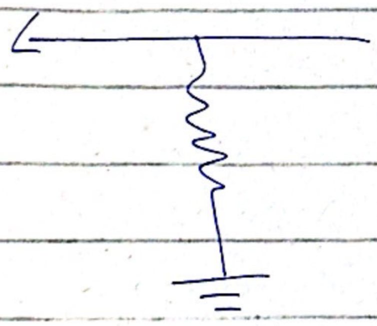
Current out = 2, 4, 8 mA
Current in = 18 mA

Pull up vs Pull down

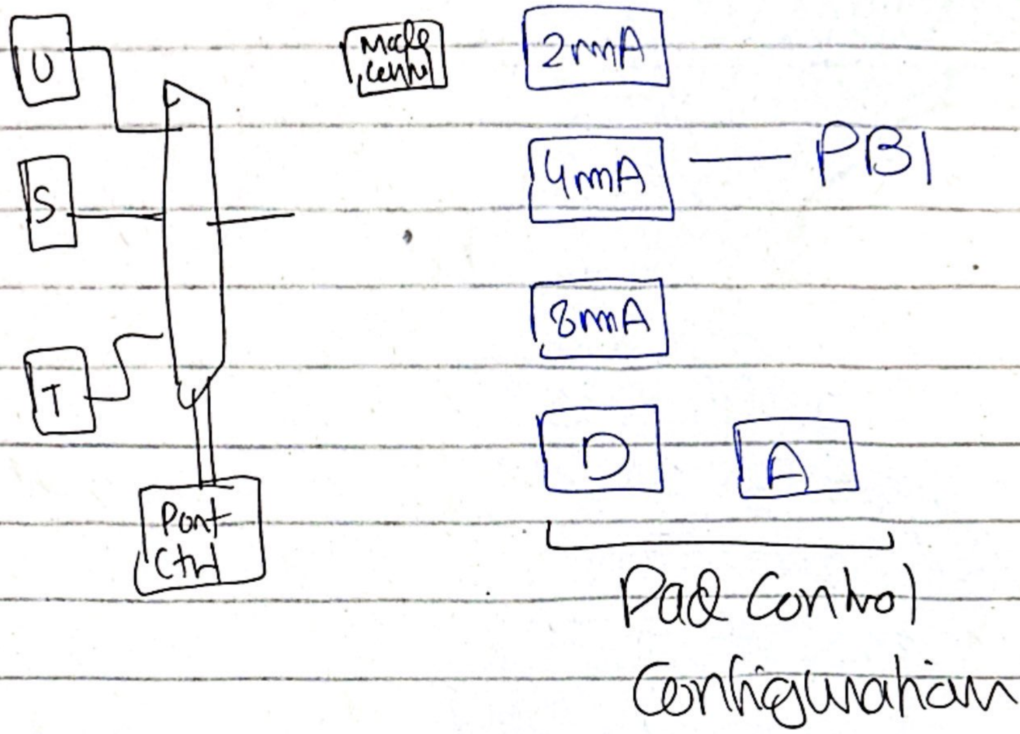
Pull up Resistor



Pull down resistor



PB7 and PF0 are by default locked



GPIO are registers
↳ require clock configuration

RCGCGPIO → Clock
↳ Base Adress: 0x400FE000
Offset: 0x608

Total address: 0x400FE608

Port A Port B Port C Port D

6 GPIOS → R0, R1, R2, R3, R4

Port E

RS

Port F

↳ 0 bit turn clock off

↳ 1 bit turn clock on

→ clock OFF ⇒ save power

PF	PE	PD	PC	PB	PA
5	4	3	2	1	0

PA

- 0 → off
- 1 → ON, Enable

Set PB → use OR $\boxed{0x02}$

```
rcgpio = rcgpio | 0x02;
```

1. Clock Control (first)

2. Mode Control
   - GPIO
   - Alternate Functionality
     - Enable Alternate Function
     - Port Control Mux
     - Analog Functionality

   $\underbrace{\text{Enable Alternate Function} \quad \text{Port Control Mux}}_{\text{Digital}}$

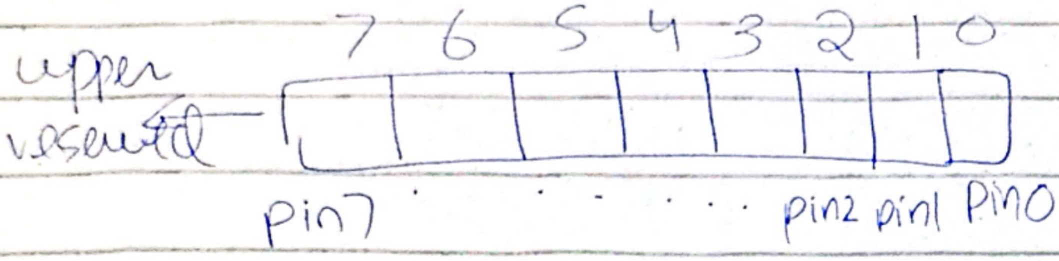
3. Pad Control

<u>GPIO portB</u>

Base: 0x40005000

AFSEE --> 0x420

GPIO AFSEEL Address: 0x40005420



0 → Disable
1 → Enable

} other than direction register

Our focus is on PB1

Enable bitset --> PB1 : OR 0x02

Disable: bit clear --> PB1 ~~0x02~~

: ~~(~A~~ AND (~0x02)

~ 00000010

$\underbrace{1111}\,\underbrace{1101}$

0XFD

By default all run on 2mA drive select

→ if 4mA: register
L
اندراج کا [illegible]
کر

steps

1. Clock enable:
   RCGCGPIO --> PortB --> 1

- - - - - - - - - - - - -

gpio internal module steps (PB1)

1. AFSEL: OFF (Alternate functionality)
   ---- GPIO PIN1

2. DEN: ON
   ---- GPIO PIN1

3. CURRENT DRIVE STRENGTH (output)
   4mA -- DR4R — ON — PIN1
   DR2R → OFF (do not)

4. Direction: Output

→ GPIODIR ⟨ 0 input / 1 output

→ GPIODATA → $\underbrace{\text{0x000}}_{\text{offset}}$ مش هنروحها

↳ we will use 0X3FC

0x20

↓

00100000